USN | | | | | | | | | | | [23CSE209]

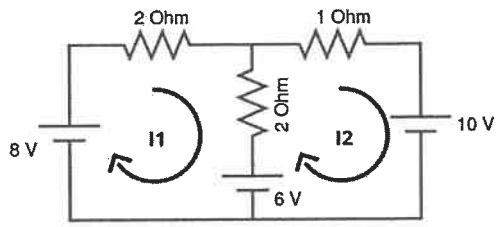
SECOND SEMESTER B. Tech EXAMINATION, JUNE - 2024

ELECTRICAL AND ELECTRONICS FOR COMPUTATIONAL THINKING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A		**05x02=10**	
1.	**a.**	An electric heater draws 3.5 A from a 110 V source. Find the resistance of the heating element.	**L3**	**C01**	**02**
	b.	Define the Frequency and Instantaneous Value of an alternating Quantity.	**L2**	**C02**	**02**
	c.	Define Continuous signal with an example.	**L2**	**C02**	**02**
	d.	Explain Feedback , Error Detector in a Control System.	**L2**	**C06**	**02**
	e.	What is Digital image processing? Mention its applications.	**L2**	**C05**	**02**
		PART - B		**06x05=3**	
2.	**a.**	By using Kirchhoff's Voltage law, find the current through the 2 Ohm resistor. 	**L3**	**C01**	**05**
		OR			
	b.	Write the difference between Passive and Active elements.	**L2**	**C01**	**05**
3.	**a.**	An alternating voltage given by V = 140 sin 314 t is connected across a pure resistor of 50 ohm. Find **i.** The frequency of the source. **ii.** The rms current through the resistor.	**L3**	**C02**	**05**
		OR			
	b.	Define RMS Value and derive an expression for the RMS value of a Sinusoidal Current.	**L2**	**C02**	**05**
4.	**a.**	Explain the even and odd components of signal with an example.	**L2**	**C02**	**05**
		OR			
	b.	Find Whether the signals are even or odd: **i.** $x(t) = e^{-5t}$ **ii.** $x(t) = \sin 2t$ **iii.** $x(t) = \cos 2t$	**L3**	**C03**	**05**
5.	**a.**	Explain the importance and advantages of grouping.	**L2**	**C04**	**05**
		OR			

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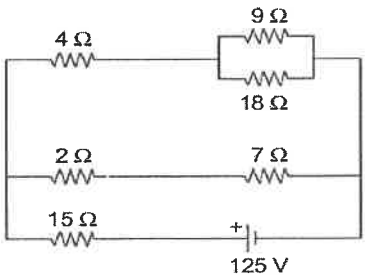
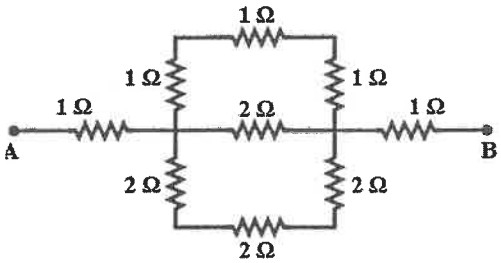
	b.	Define Signal processing and explain the need of Pre- processing of a signal.	L2	C04	05
6.	a.	Define Open loop Control system. Explain the advantages, disadvantages of Open loop Control system.	L2	C06	05
		OR			
	b.	Write the Difference between the Open loop and Closed loop control systems.	L2	C06	05
7.	a.	Explain the fundamental steps in digital image processing.	L2	C05	05
		OR			
	b.	List the application of Digital Image Processing.	L2	C05	05
		PART - C			06x10=60
8.	a.	Find i. Current in 15 Ω resistor, ii. Voltage across 18Ω resistor, and iii. Power dissipated in 7Ω resistor of the circuit given in the figure  	L3	C01	10
		OR			
	b.	Calculate the effective resistance between the points A and B in the circuit shown in figure  	L3	C01	10
9.	a.	A coil of resistance 10 Ω and inductance 0.1 H is connected in series with a 150 μF capacitor across 200 V, 50 Hz supply. Calculate i. Inductive reactance, capacitive reactance, impedance, current and power factor ii. The voltage across the coil and capacitor respectively.	L3	C02	10
		OR			
	b.	Derive the expression for voltage, current and power for a single-phase AC circuit containing R and L in series. Draw the necessary waveform and phasor diagrams.	L2	C02	10
10.	a.	Define signal, system and explain the classification of signals.	L2	C03	10
		OR			

	b.	Explain the Pre- processing of a signals and write the different types of pre- processing techniques.	**L2**	**C03**	**10**
11.	**a.**	Explain the Power and Energy signals with an example.	**L2**	**C04**	**10**
		OR			
	b.	Write the Difference between the Analog and Digital signals.	**L2**	**C04**	**10**
12.	**a.**	Explain the fundamental steps used in Digital Image Processing.	**L2**	**C05**	**10**
		OR			
	b.	Elaborate the different components used in Image Processing System.	**L2**	**C05**	**10**
13.	**a.**	Describe the design of control system for electrical clothes Drier.	**L2**	**C06**	**10**
		OR			
	b.	Explain the design of Control system for air conditioning system.	**L2**	**C06**	**10**